An Introduction to Federated Learning

Yuchang SUN
2022-02-15

Hello!

I am Yuchang Sun

Year-1 PhD student in ECE at HKUST

Supervised by Prof. Jun ZHANG

Research interest: Efficiency of FL

Find me at https://hiyuchang.github.io

Outline

- Introduction
- Federated Learning
- Applications
- Bottlenecks in Efficiency
- Useful Tools for You

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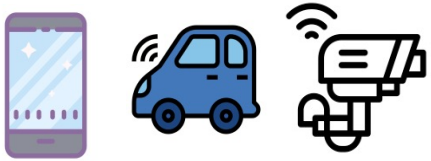
Introduction

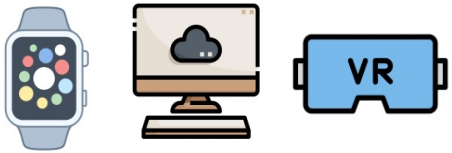
Let’s go back where we start.

The Success of AI, IoT, VR, ...

Generating data



Promoting AI



Data Silos of Hospitals, Banks, ...

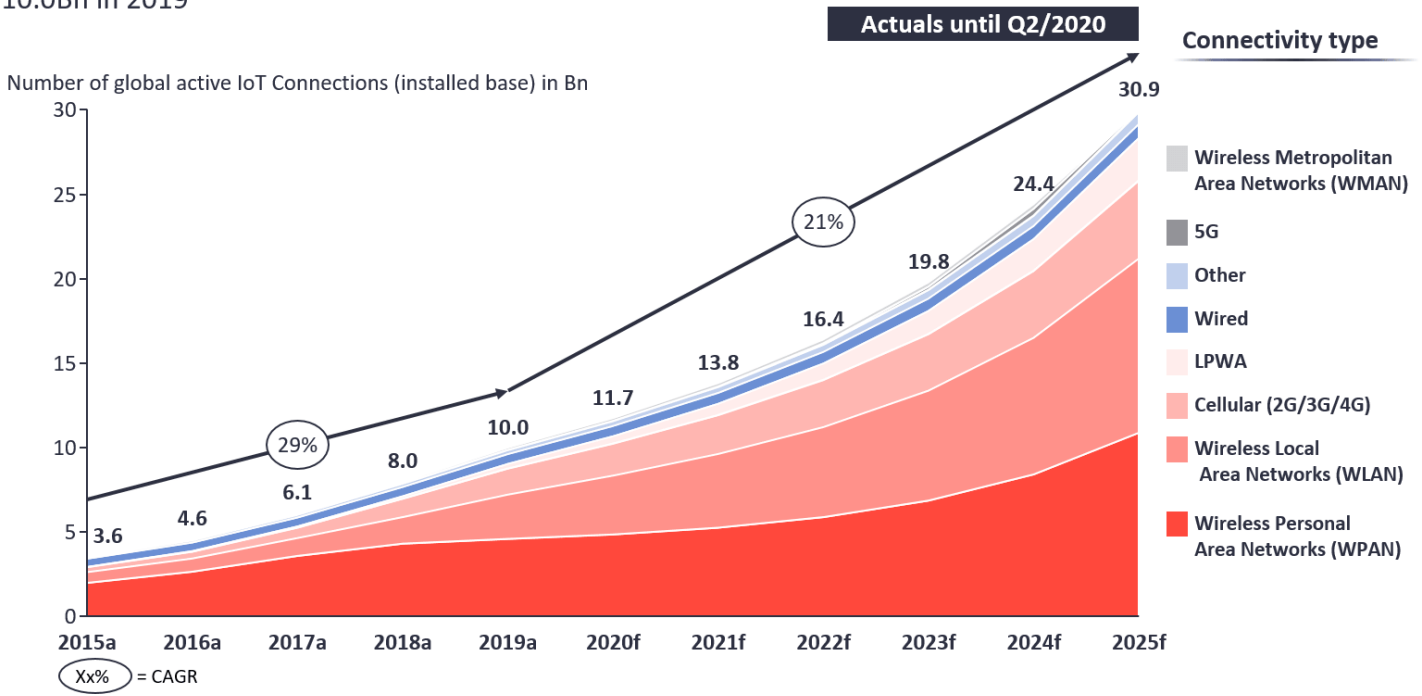
Generating data

User Analysis

Privacy-sensitive data

Global Number of Connected IoT Devices

10.0Bn in 2019



Note: IoT Connections do not include any computers, laptops, fixed phones, cellphones or tablets. Counted are active nodes/devices or gateways that concentrate the end-sensors, not every sensor/actuator. Simple one-directional communications technology not considered (e.g., RFID, NFC). Wired includes Ethernet and Fieldbuses (e.g., connected industrial PLCs or I/O modules); Cellular includes 2G, 3G, 4G; LPWAN includes unlicensed and licensed low-power networks; WPAN includes Bluetooth, Zigbee, Z-Wave or similar; WLAN includes Wi-fi and related protocols; WMAN includes non-short range mesh, such as Wi-SUN; Other includes satellite and unclassified proprietary networks with any range.

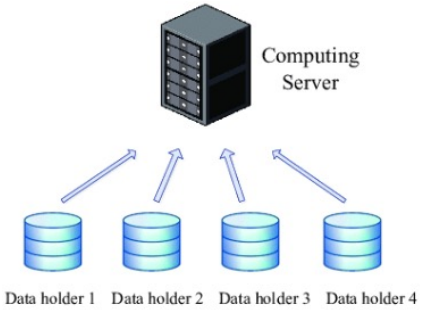
Source(s): IoT Analytics - Cellular IoT & LPWA Connectivity Market Tracker 2010-25

How to utilize these data?

Traditional Approaches

Centralized Training

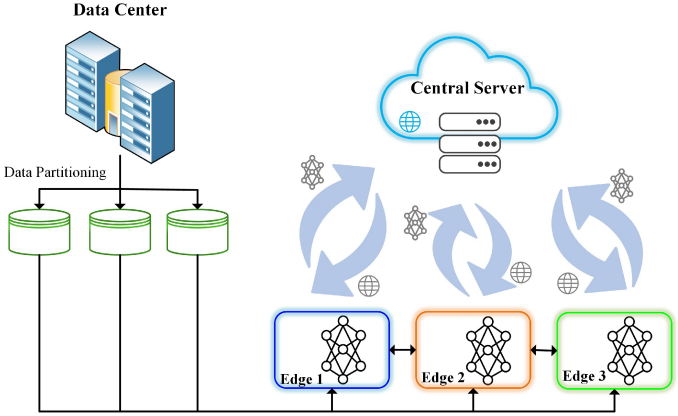
- Upload the data to the server.
- Let server train the model.



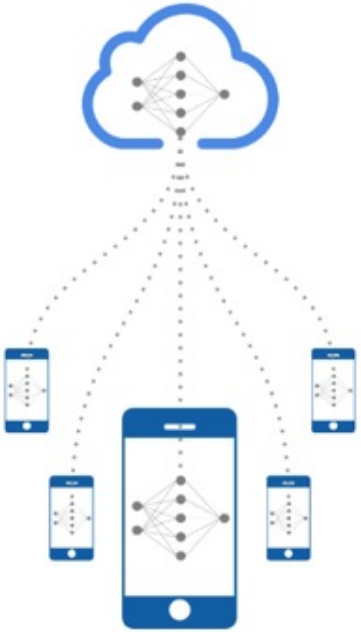
Privacy leakage & high latency!

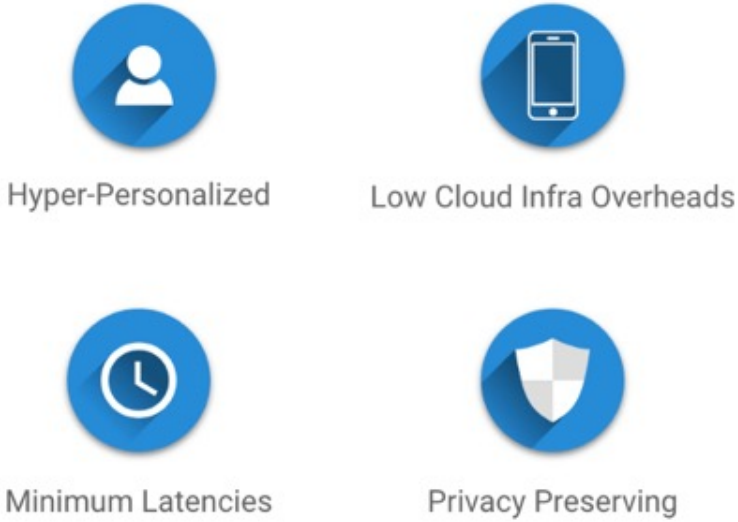
Distributed Learning

- The server distribute the computing tasks to the devices.



Federated Learning Emerges!





[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in Proc. Int. Conf. Artif. Intell. Statist. (AISTATS), Ft. Lauderdale, FL, USA, Apr. 2017.

Federated learning aims to train a single model from multiple data sources, under the constraint that data *stays at the source.*

“

2 Federated Learning

How does FL work?

Training Process

(1) Model Broadcasting

The centralized server broadcasts the model to available devices.

Source: https://federated.withgoogle.com

Training Process

(2) Local Training

The device trains the model based on its data.

Training Process

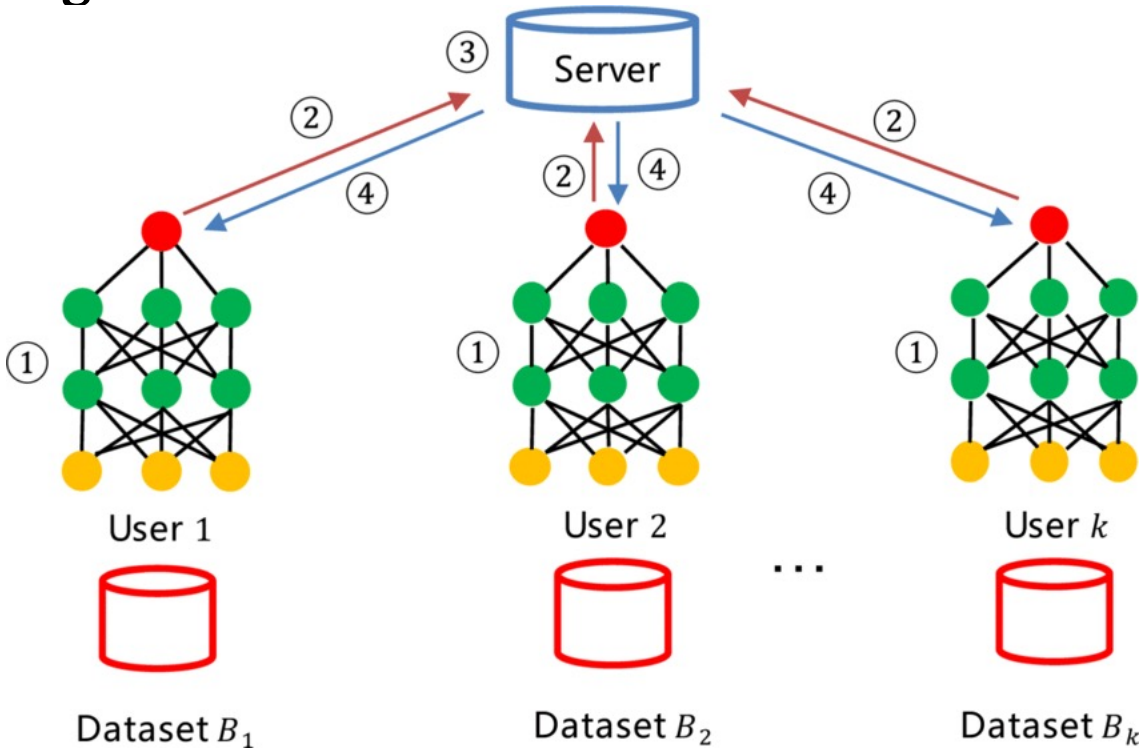
(3) Model Aggregation

The device uploads result/model/updates to the server.

The server aggregates these models.

Data stays locally!

Training Process



Categories (1)

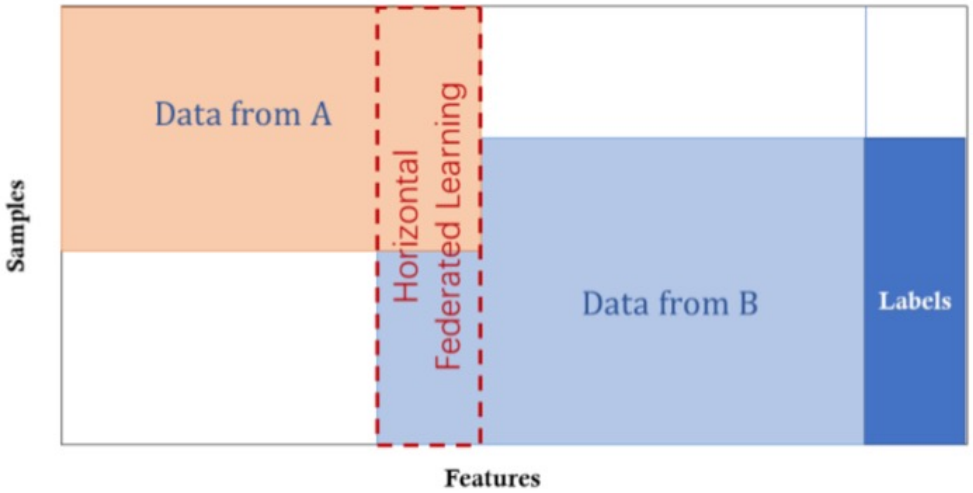
Horizontal FL
Divided by samples

Vertical FL
Divided by features

Federated transfer learning

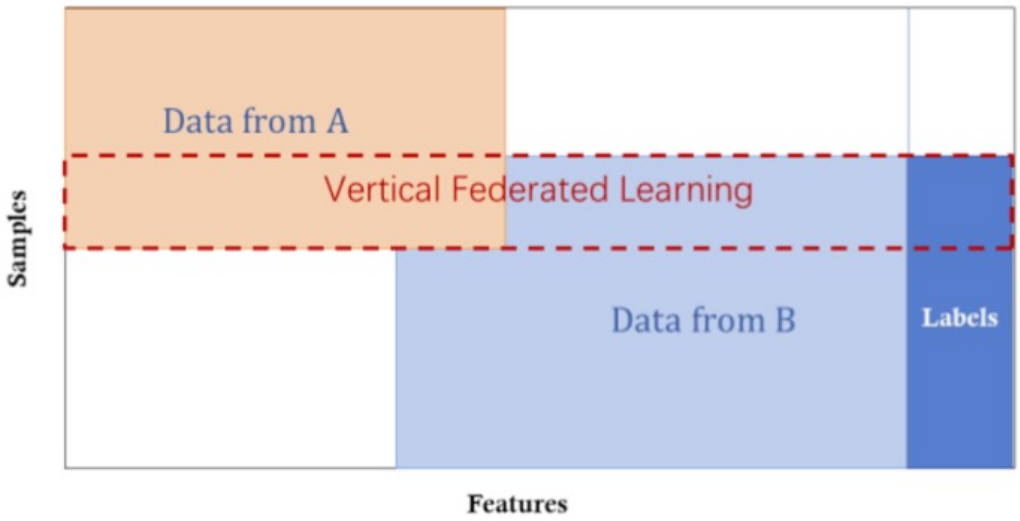
Categories (1)

Horizontal FL
Divided by samples



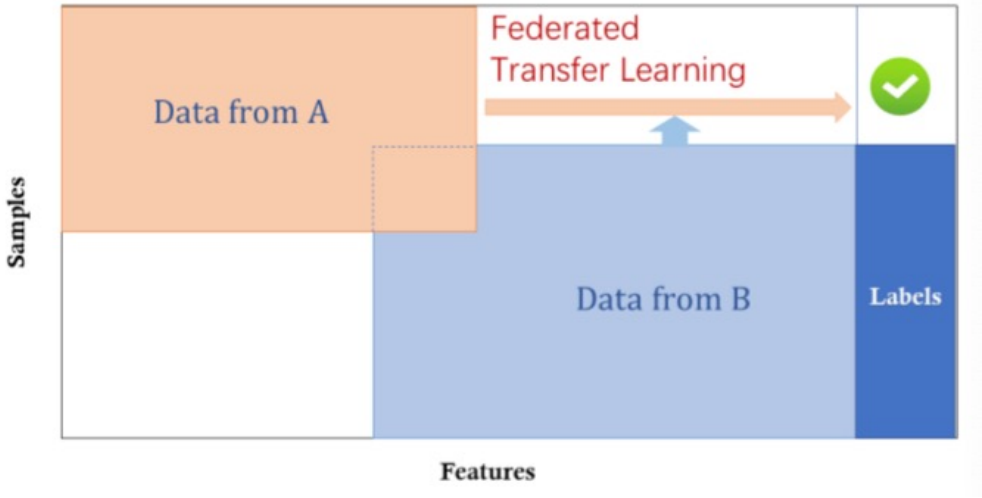
Categories (1)

Vertical FL
Divided by features



Categories (1)

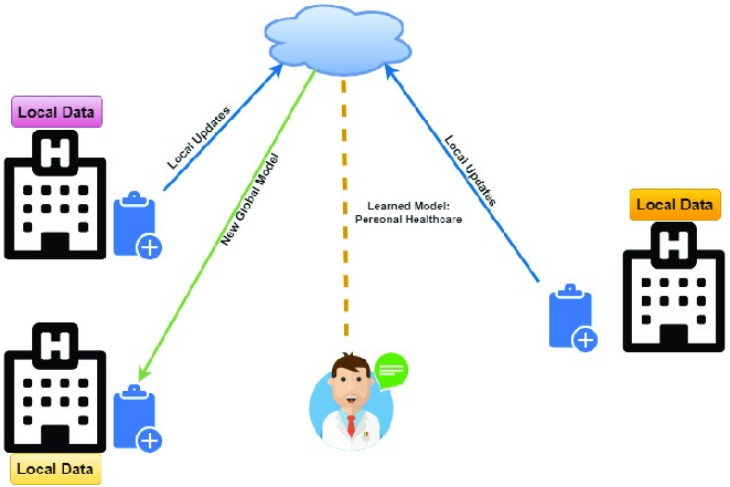
Federated transfer learning



Categories (2)

Cross-device FL

Cross-silo FL



3

Applications

Why does FL help?

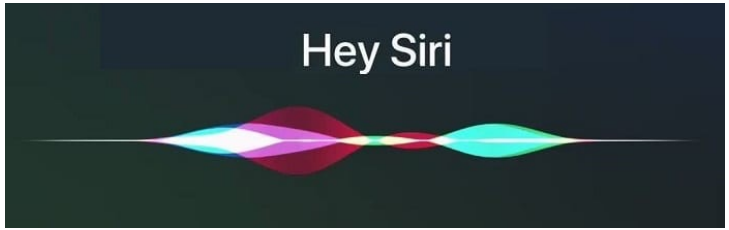
The future of AI is on the Edge.

Privacy preserving

Anytime anywhere

Applications

- Hey Siri!
- The raw audio of users' Siri requests never leaves their iPhones and iPads, but the assistant continuously gets better at identifying the right speaker.



- Webank AI

[2] Paulik M, Seigel M, Mason H, et al. Federated evaluation and tuning for on-device personalization: System design & applications[J]. arXiv preprint arXiv:2102.08503, 2021.
[3] https://cn.fedai.org

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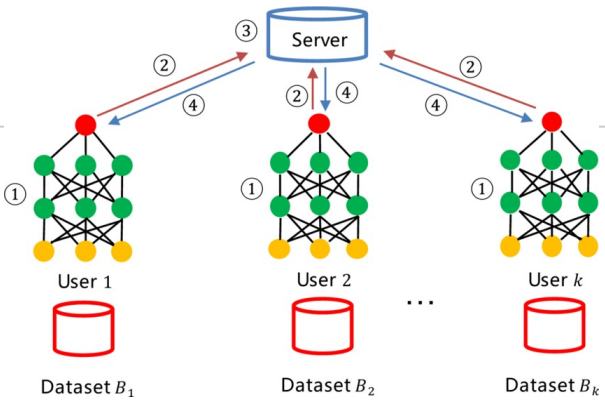
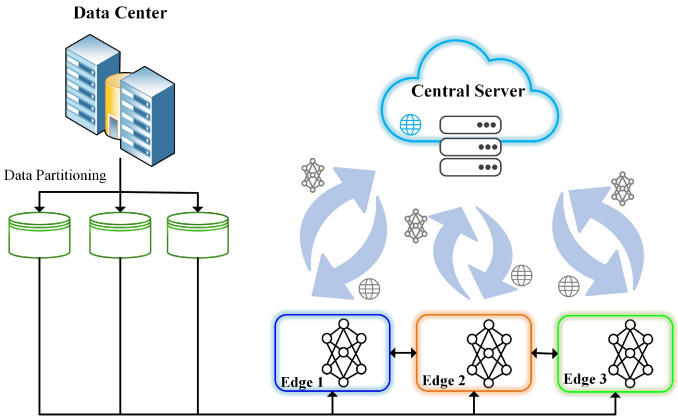
Bottlenecks in Efficiency

What's new challenges?

Different from Distributed Learning!

- Raw data never leaves the device.
- The data is NOT distributed by the server.
- FL generally leverages encryption techniques.

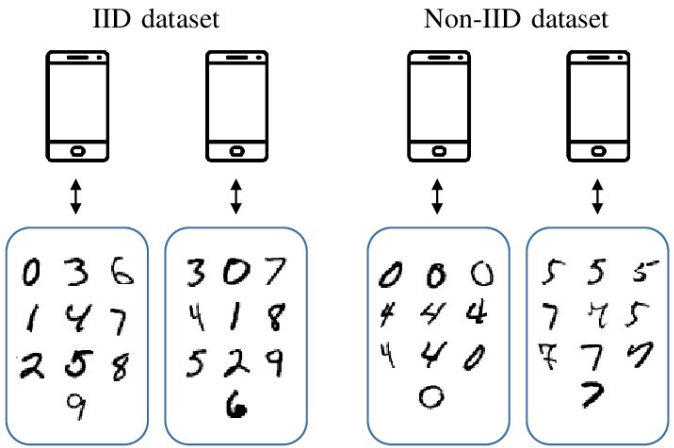


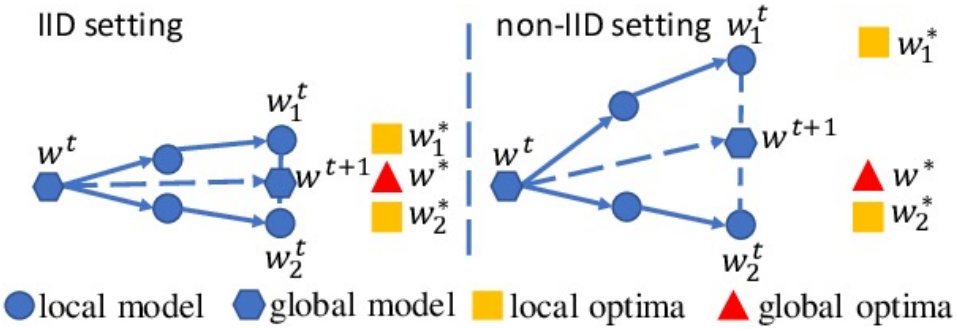


Non-IID Data

The data is born at the devices.

They may follow different distributions.

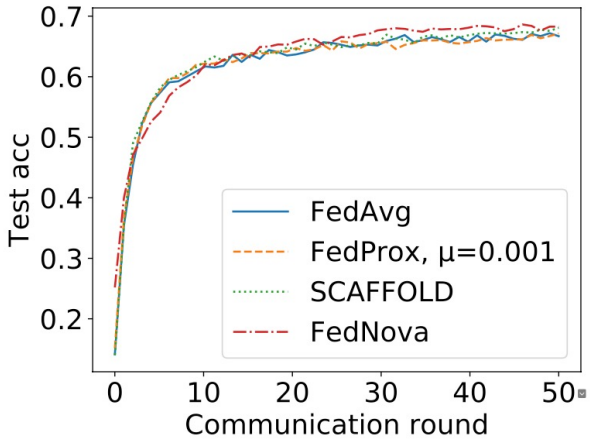




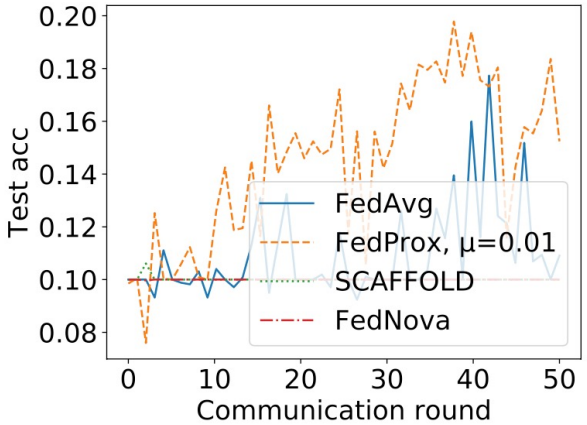
Non-IID; Non-identical and independently distributed

Non-IID Data

But this significantly degrades the training performance!



(a) $p_k \sim Dir(0.5)$



(b) $\#C = 1$

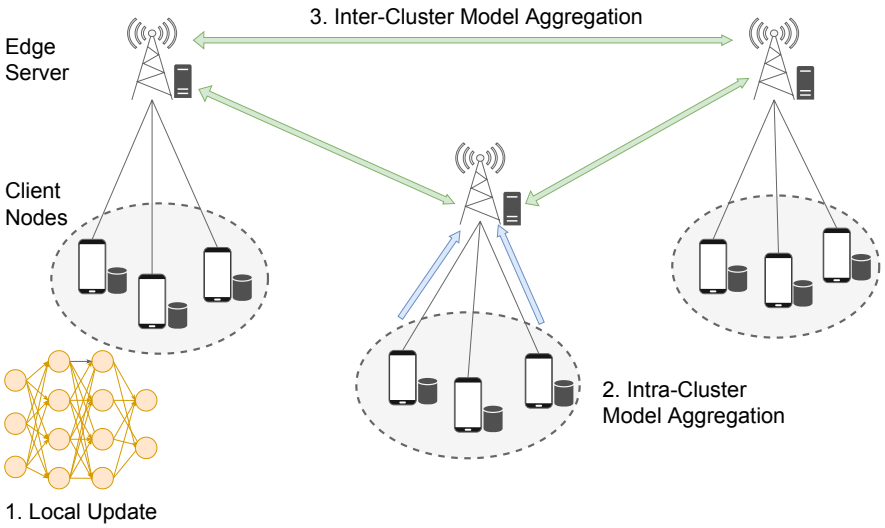
NOTE: CNN on CIFAR-10 dataset

[4] Li Q, Diao Y, Chen Q, et al. Federated learning on non-iid data silos: An experimental study[J]. arXiv preprint arXiv:2102.02079, 2021.
[5] Li X, Huang K, Yang W, et al. On the convergence of fedavg on non-iid data[J]. arXiv preprint arXiv:1907.02189, 2019.

Communication latency

The high latency with cloud server

- Compression
- New architecture

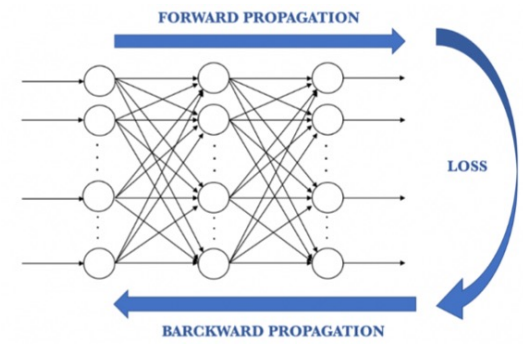


[6] Y. Sun, J. Shao, Y. Mao, J. H. Wang, and J. Zhang, “Semi-decentralized federated edge learning for fast convergence on non-IID data,” accepted to IEEE Wireless Commun. Netw. Conf., Austin, TX, USA, Apr. 2022.

Straggler Effect

Local training

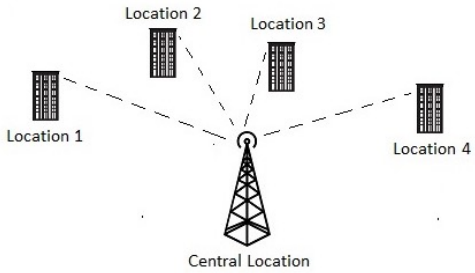
Dozens of GFLOPs



Diverse computational resources

Model uploading

Millions of parameters

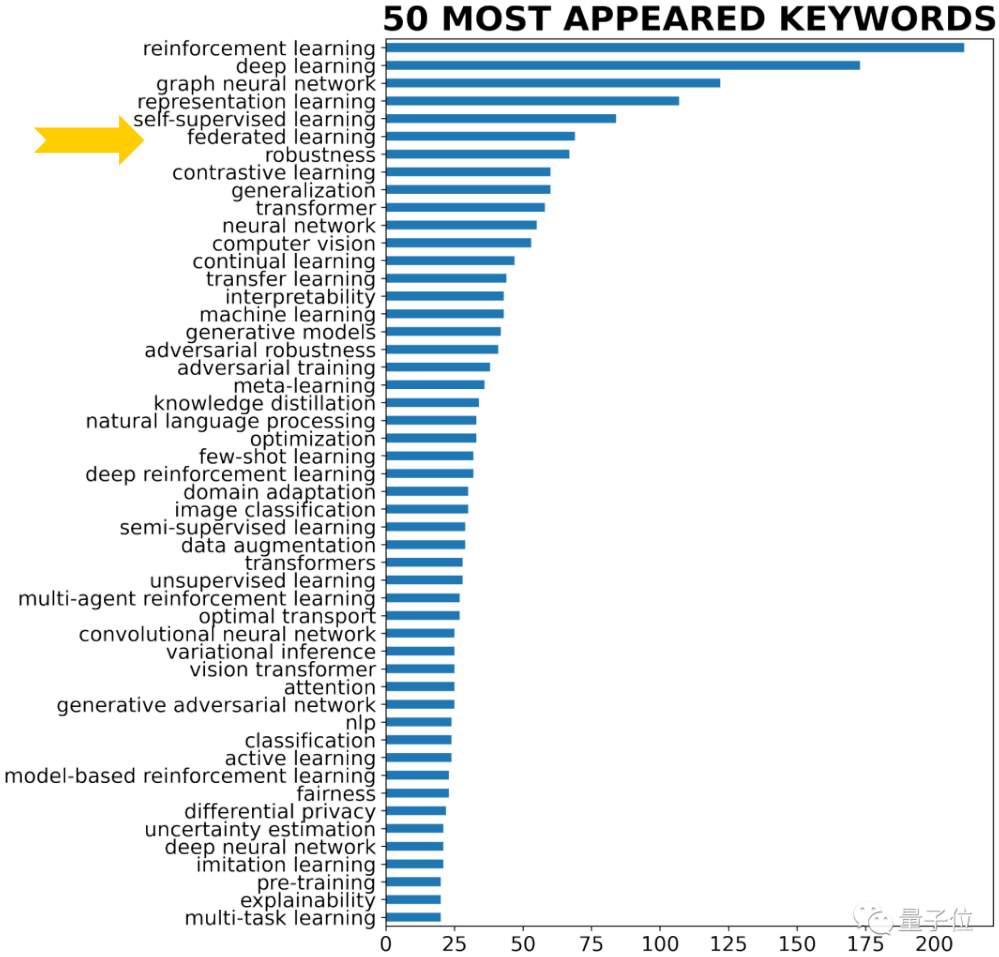


Different channel states

Device Dropout

Worse more, the device may drop out anytime.

Much worse than stragglers!



Top 7 in ICLR 2022

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Useful Tools for You

Wanna start?

Useful Tools/Links/Repos

Surveys

Problems [7]
Optimization [8]
FEEL [9]

Frameworks

FedML
FATE
LEAF
Flower
...

GitHub Repos

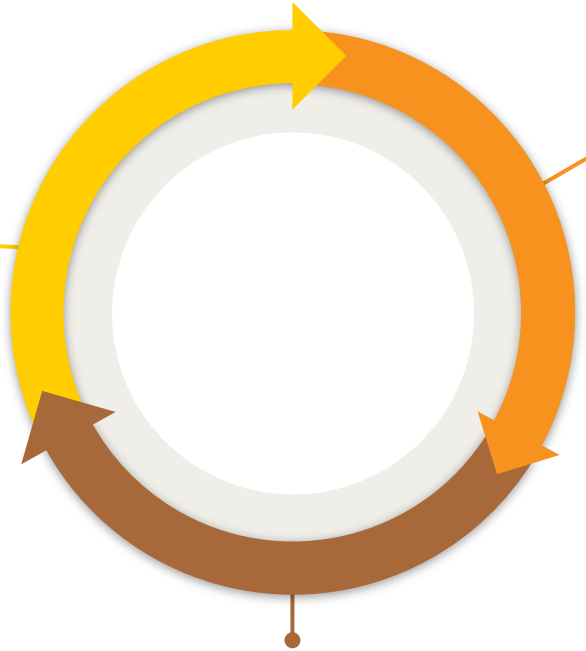
“Awesome federated learning”

[7] McMahan H B. Advances and open problems in federated learning[J]. Foundations and Trends® in Machine Learning, 2021, 14(1).
[8] Wang J, Charles Z, Xu Z, et al. A field guide to federated optimization[J]. arXiv preprint arXiv:2107.06917, 2021.
[9] W. Y. B. Lim, N. C. Luong, D. T. Hoang, Y. Jiao, Y.-C. Liang, Q. Yang, D. Niyato, and C. Miao, “Federated learning in mobile edge networks: A comprehensive survey,” IEEE Commun. Surveys Tuts., vol. 22, no. 3, pp. 2031–2063, 3rd Quart. 2020.

Summary

Main Takeaway Today

Federated learning makes privacy-preserving training possible.



FL distinguishes from distributed learning in data distribution and hardware requirements.

There are still many challenges and open problems in FL.

Thanks!

Any questions ?

You can find me at

- https://hiyuchang.github.io
- yuchang.sun@connect.ust.hk